

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of M. Sc. (AOC) Examination May 2018

OC 829: Synthetic Dyes and Pigments

Date: 01/05/2018 Day: Tuesday Time: 10.00 a.m. to 10.30 a.m. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
-

Q-I A Choose the correct answer for the following questions. 10

1 The chromophore in benzene azo p-cresol is

- A. nitroso group
- B. azoxy group
- C. azo group
- D. none of the above

2 Bromamine acid contains

- A. -NH_2 and -OH group
- B. -NH_2 , $\text{-SO}_3\text{H}$ and -Br group
- C. -NH_2 and -COOH group
- D. None of the above

- 3 The auxochromes in benzene azo resorcinol is
- 2OH group
 - N=N group
 - N=O group
 - A and B
- 4 Visible range corresponds to
- below 35.5 Kcal.mole⁻¹
 - 150-250 Kcal.mole⁻¹
 - 35.5-71.0 Kcal.mole⁻¹
 - 71.0-150 Kcal.mole⁻¹
- 5 Examples for strong chromophoric groups are
- C=C, C=O, C=S
 - NH₂, NHR, NR₂, OH
 - NO₂, N=O, N=N
 - None of the above
- 6 The λ_{max} of 1,5-diNH₂ AQ dyes is similar to
- 2-NH₂AQ
 - 2,5-diNH₂AQ
 - 1-NH₂AQ
 - 3-NH₂AQ
- 7 The chromophore in methyl red dye:
- Azoxy group
 - Azo group
 - Nitroso group
 - A and B
- 8 As the number of fused ring increases the colour deepens.
- Not effected
 - More (from colourless to black)
 - Less
 - To yellow

9 Methyl red used as

- A. laxative
- B. indicator in acid-base titrations
- C. antitumor agents
- D. none of the above

10 Reactive dyes react with substrate by forming a

- A. hydrogen bond
- B. ionic bond
- C. Π -bond
- D. σ -bond

Q-I B Draw the structure of the following dyes.

10

1 Malachite green

2 Polar yellow brown

3 Rhodamine

4 Iron green lake of fast green O

5 Naphthol blue black 6B

6 Auramine G

7 Fluorescein

8 Primuline base

9 Amido yellow E

10 Croconium dyes

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of M. Sc. (AOC) Examination May 2018

OC 829: Synthetic Dyes and Pigments

Date: 01/05/2018 Day: Tuesday Time: 10.30 a.m. to 01.00 p.m. Maximum Marks: 50

Important Instructions:

- Section I and II must be attempted in TWO ANSWER SHEET.
 - Make suitable assumptions and draw neat figures wherever required.
 - Use of non-programmable calculator is allowed.
 - Show necessary calculations.
-

SECTION-I

Q-II Answer the following. 10

- A** Mention the factors which are important in the reactive dyes. 3
- B** Explain: Oxidative hair dyes. 3

OR

- B** Mention the characteristics of Fluorescence Brightener. 3
- C** Mention the terms involved in FBA and explain their role. 4

OR

- C** Give the properties of Tinopal CBS-X and Tinopal OB. 4

Q-III Answer the following. 10

- A** Mention the general approach to produce NIR dyes giving examples. 4
- B** Why 1,4 – Donar AQ is more bathochromic than 1,5 – Donar AQ. 3
- C** Explain why 1–Donar AQ isomer of AQ is more bathochromic than 2–Donar AQ. 3

OR

- C** Write note on Cyanine type chromophore. 3

SECTION-II

Q-IV Answer the following. 10

- A** Explain the effect of solvent on tautomerism. (in case of azo dyes) 3
- B** Write note on Polyazodyes in which azo group is not conjugated to each other. 3

OR

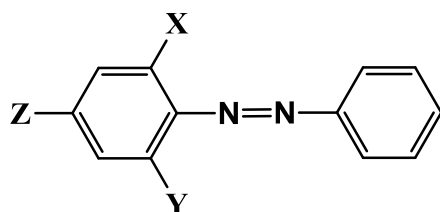
- B** Explain the tautomerism of hydroxyazo dyes of benzene series. 3
- C** Why tautomerism is not observed in α -phenyl azo-phenol dyes but observed in similar naphthol derivative. 4

Q-V Answer the following.**10****A** Write note on: chromophore theory.

3

B

3



Sr. No	X	Y	Z	$\lambda_{\max}$ (nm)
1	H	H	NO ₂	453
2	NO ₂	H	H	425
3	NO ₂	H	NO ₂	491
4	NO ₂	NO ₂	NO ₂	520

Explain the above data.

C Mention the characteristics of the dye.

4

Q-VI Answer the following.**10****A** How one can get yellow dyes in colour photography.

3

B Explain: "Brooker Deviation" giving suitable example.

3

C Explain: the dye forming process in colour photography.

4

OR**C** Write note on: The image diffusion process.

4